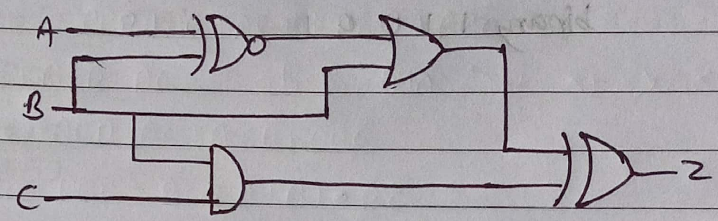
Assignment 2
Boolean Algebra

1) Derive the output expression 'Z' for the below circuits and draw the simplified circuits.

(a)



→ From the circuit

- First Gate = XNOR
  $(A \oplus B)' = AB + A'B'$
- second gate = OR
  $(A \oplus B)' + B$
- Bottom Gate = AND
  BC
- Final Gate = XOR

Output expression : $f = ((A \oplus B)' + B) \oplus (BC)$

Simplify:

$AB + B = B(A+1) = B$

$(A \oplus B)' = AB + A'B'$

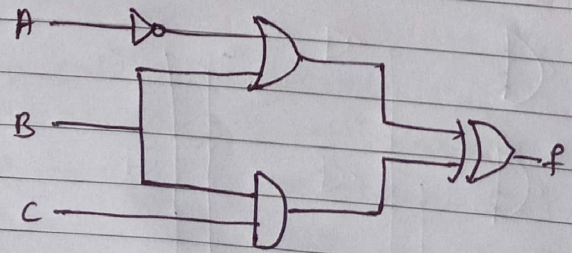
$B + A'B' = (A' + B)(B + B')$

$\underline{AB} + A'B' + \underline{B} = B + A'B' = B + A'$

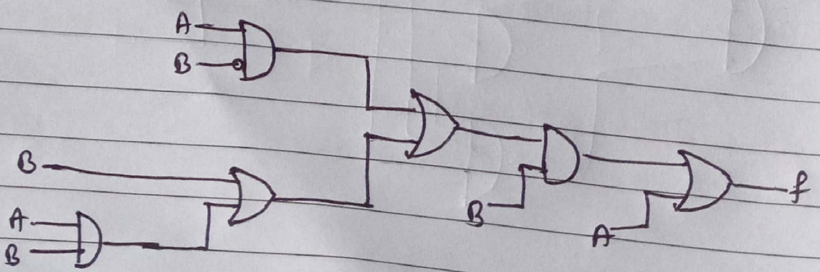
$= \underline{\underline{A' + B}}$

∴ simplified Expression:

$f = (A' + B) \oplus BC$



(b)



→ from the circuit,

- TOP AND : AB'
- Bottom AND : AB
- First OR : $B+AB=B$
- second OR : $AB'+B=A+B$
- AND with B :

$(A+B)B = AB+BB = AB+B = B(A+1) = B$

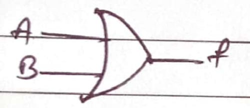
- last OR :

$B+A = A+B$

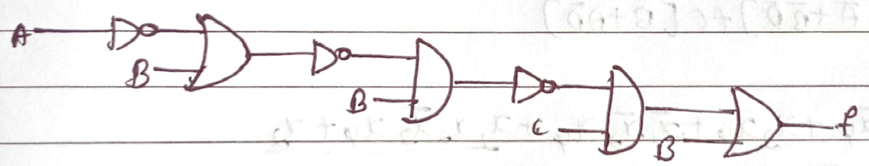
output expression : $f = [(AB')+(B+AB)]B+A$

simplified expression:

$f = A+B$



(c)



→ from the circuit

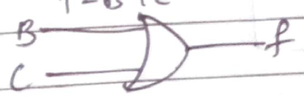
- NOT on A : $\bar{A}$
- OR : $\bar{A}+B$
- NOT : $\overline{(\bar{A}+B)} = \bar{\bar{A}}\,\bar{B} = A\bar{B}$
- AND : $A\bar{B}B = 0$
- NOT : $\bar{0} = 1$
- AND : $1.C = C$
- OR : $B+C$

o/p expression : $f = [\overline{(\overline{(\bar{A}+B)}B)}C]+B$

$f = [((A'+B)'B)'C]+B$

Simplified expression

$f = B+C$



2] minimize the following Boolean expressions using Boolean postulates.

(a) $\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

→ $\bar{A}\bar{D}(CE+\bar{B}) + AB(CE+D)$

(b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

→ · $\bar{A}C + \bar{A}C\bar{D} = \bar{A}C$ absorption law: $X + XY = X$

$\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D + A\bar{B}\bar{C}$

· $\bar{A}\bar{B}C + A\bar{B}\bar{C} = \bar{B}\bar{C}(\bar{A}+A) = \bar{B}\bar{C}$

$\bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D$

$ABD + \bar{A}C + \bar{C}(\bar{B} + AD)$

(c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

→ · $\bar{A}BC + BC = BC$ absorption law: $X+XY=X$

$\bar{A}\bar{C} + BC + A\bar{B} + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

$\bar{C}[\bar{A} + \bar{B}\bar{D}] + BC + A\bar{B} + AC\bar{D}$

$A\bar{B} + \bar{C}[\bar{A} + \bar{B}\bar{D}] + C[B + A\bar{D}]$

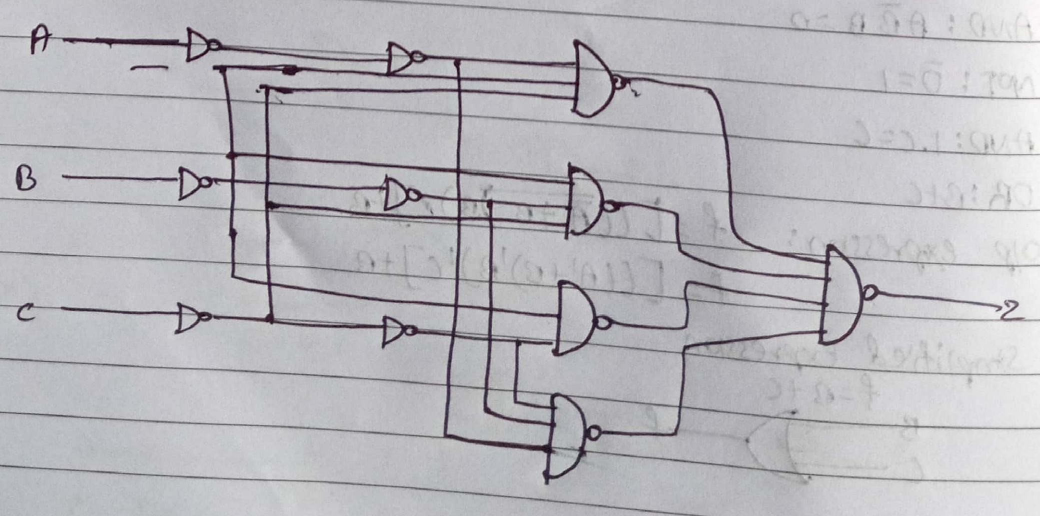
(d) $\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$

· $x_2 + x_1x_2\bar{x}_3x_4 = x_2$ absorption law: $x + xy = x$

$\bar{x}_1\bar{x}_2\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_2$

$x_1[\bar{x}_3\bar{x}_4 + \bar{x}_2x_4] + x_3x_4 + x_2$

3] which single gate can represent the functionality of given circuit?



tx - T2
rx - T3

→ The first 4 Nand gates give 4 minterms corresponding to the cases where an odd number of inputs are 1.
• $\bar{A}, \bar{B}, C$ • $\bar{A}, B, \bar{C}$ • $A, \bar{B}, \bar{C}$ • A, B, C
The final Nand combines these o/ps giving
$Z = \overline{\bar{P_1}, \bar{P_2}, \bar{P_3}, \bar{P_4}} = P_1 + P_2 + P_3 + P_4$
$Z = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$
This is exactly the boolean expression for a 3-i/p exclusive-OR-(XOR) gate:
$Z = A \oplus B \oplus C$

4) minimize the following Boolean expressions using k-map
(a) $f(a,b,c,d) = \Sigma m\,(4,6,8,10,11,12,15) + \Sigma d\,(3,5,7,9)$
→

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
00 $\bar{A}\bar{B}$	(0)	(1)	X (3)	(2)
01 $\bar{A}B$	1 (4)	X (5)	X (7)	1 (6)
11 AB	1 (12)	(13)	1 (15)	(14)
10 $A\bar{B}$	1 (8)	X (9)	1 (11)	1 (10)

$\therefore f = \bar{A}B + A\bar{B} + CD + A\bar{C}\bar{D}$

(b) $f(a,b,c,d) = \pi M\,(0,2,4,6,7,9) + \pi D\,(10,11)$

AB \ CD	$C+D$ 00	$C+\bar{D}$ 01	$\bar{C}+\bar{D}$ 11	$\bar{C}+D$ 10
00 $A+B$	0 (0)	(1)	(3)	0 (2)
01 $A+\bar{B}$	0 (4)	(5)	0 (7)	0 (6)
11 $\bar{A}+\bar{B}$	(12)	(13)	(15)	(14)
10 $\bar{A}+B$	(8)	0 (9)	X (11)	X (10)

$\therefore f = (A+D)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{D})$